

AMOGH MANNEKOTE

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PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Science at UF, specializing in Conversational AI and Generative AI. Research focuses on human behavior simulation, LLM alignment, synthetic data generation, and evals, supported by extensive experience in machine learning, natural language processing, and software engineering. Possesses a proven track record of research publications and industry experience at organizations including Amazon Alexa, Rasa, and Openstream—particularly on task-oriented dialogue agents. Dedicated to developing advanced conversational AI solutions that enhance user experience.

EDUCATION

University of Florida	Gainesville, FL
PhD in Computer Science	01/2021 - Present
Ramaiah Institute of Technology	Bengaluru, India
B.E. in Computer Science & Engineering	08/2014 - 08/2018

PUBLICATIONS AND PREPRINTS

1. **Amogh Mannekote**, A. Davies, J. Kang, K.E. Boyer (2025). [Can LLMs Reliably Simulate Human Learner Actions? A Simulation Authoring Framework for Open-Ended Learning Environments](#). *To Appear in AAAI Proceedings*.
2. X. Tian, **Amogh Mannekote**, C. E. Solomon et al. (2024). [Examining LLM Prompting Strategies for Automatic Evaluation of Learner-Created Computational Artifacts](#). *In Educational Data Mining (2024)*.
3. **Amogh Mannekote**, J. Nam, Z. Li, J. Gao, K.E. Boyer, B. Dorr (2024). [IndirectRequests: Making Task-Oriented Dialogue Datasets More Natural by Synthetically Generating Indirect User Requests](#). *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*.
4. **Amogh Mannekote**, X. Tian, K.E. Boyer, and B. Dorr (2024). [Can Similarity-Based Domain-Ordering Reduce Catastrophic Forgetting for Intent Recognition?](#) (arXiv:2402.14155)
5. **Amogh Mannekote**, M. Celepkolu, J.B. Wiggins, and K.E. Boyer (2023). [Exploring Usability Issues in Instruction-Based and Schema-Based Authoring of Task-Oriented Dialogue Agents](#). *In ACM conference on Conversational User Interfaces (CUI '23)*.
6. **Amogh Mannekote**, B. Dorr, K.E. Boyer (2023). [Agreement Tracking for Multi-Issue Negotiation Dialogues](#) (arXiv:2307.06524)
7. **Amogh Mannekote** (2023). [Towards a Neural Era in Dialogue Management for Collaboration: A Literature Survey](#) (arXiv:2307.09021)

WORK EXPERIENCE

- Amazon Alexa**, *Research Intern* Bellevue, WA (09/2024 - 12/2024)
- Developed MENTOR, a framework for evaluating LLMs on diverse in-context feedback and tool use.
 - Conducted experiments on LLM adaptability to structured feedback in programming and reasoning tasks.
- Openstream.ai**, *Research Intern* Remote (Sep 2023 – Dec 2023)
- Developed a LangChain-based evaluation pipeline for semantic parsing for a neurosymbolic dialogue system, improving model evaluation efficiency by 30%.
- Amazon Alexa**, *Research Intern* Bellevue, WA (May 2023 – Sep 2023)
- Designed and implemented a pipeline for synthetic generation of indirect utterances in task-oriented dialogue using linguistic attributes, increasing dataset diversity by 25%.
- University of Florida**, *Graduate Research Assistant* Gainesville, FL (Jan 2021 – Present)
- Created a conversational learning partner within [EarSketch](#), facilitating programming education through music remixing for high school students.
- Fractal Analytics & Crux Intelligence**, *Data Scientist* Bengaluru, India (Jan 2020 – Nov 2020)

- Implemented an end-to-end evaluation framework for Text-to-SQL semantic parsing, reducing time-to-release from one day to a few minutes.

Rasa, *Machine Learning Solutions Architect*

Berlin, Germany (Jun 2019 – Nov 2019)

- Contributed to Rasa's open-source NLU and dialogue management modules.
- Developed conversational AI solutions for enterprise clients in healthcare, automotive, insurance, and retail sectors.

Indian Institute of Science, *Research Assistant*

Bengaluru, India (07/2018 - 06/2019)

- Extensively studied and evaluated various state-of-the-art approaches to Visual SLAM with sensor fusion in challenging outdoor environments
- Improved the accuracy of an extrinsic calibration system that was based on Perspective-n-Point (PnP) by introducing a Grid Search step over the pixel-space of the images

IBM, *Intern — Watson Machine Learning Team*

Bengaluru, India (Apr 2018 – Jun 2018)

- Improved deployment efficiency by working on the serving infrastructure for TensorFlow, Keras, and PyTorch models on IBM Watson's ML platform.

Stride.AI, *Research Intern*

Bengaluru, India (Feb 2017 – Feb 2018)

- Designed a multi-turn conversational information-retrieval agent to automate employee skill matching for a Fortune 500 organization.
- Evaluated various NLP approaches including topic extraction, semantic similarity, and Siamese networks.

Indian Institute of Science (Dept. of CSA), *Intern — Game Theory Lab*

Bengaluru, India (Jun 2017 –

Aug 2017)

- Simulated prediction markets using sentiment analysis of news sources to model market behaviors.

SELECT PROJECTS

LLM4Qual LLM4Qual is a unified library that combines the standardization of HuggingFace Datasets with the power of LangChain to act as a stand-in for human annotators in text-annotation tasks. By leveraging LLMs, LLM4Qual provides a sophisticated and automated approach to evaluating qualitative data across multiple dimensions. This project is targeted at resource-poor academic laboratories that cannot afford expensive proprietary subscriptions and for developers handling sensitive data.

MultiWOZ-Gaze Although eye-gaze has been used extensively in NLP, it is underexplored in the context of dialogue corpora. This proof-of-concept demonstrates the feasibility of using inexpensive eye-tracking equipment to derive key insights about the MultiWOZ corpus of task-oriented dialogue.

Nonechucks An open-source library that provides wrappers around PyTorch's DataLoaders to allow for seamless handling of invalid or unwanted data samples in messy, real-world ML data pipelines (featured on GitHub's Trending Python Repositories).

SKILLS

Programming	Python, Java, Javascript, TypeScript, Erlang, SQL, $\text{\LaTeX}$
Machine Learning	PyTorch, TensorFlow, Keras, NumPy, Pandas, scikit-learn, Matplotlib, WandB
NLP	HuggingFace Transformers, spaCy, NLTK, Amazon Lex, Rasa
Infrastructure	AWS, GCP, SLURM, Kubernetes, Docker, Git, Bash
Web Technologies	ReactJS, HTML, Angular, MVC, Flask, NodeJS, CSS

TALKS & PODCASTS

Guest Speaker on the CS-Ed Podcast	On the use of discussion forums in CS1/2 classrooms [Link]
Building a Chatbot with Rasa	Delivered at Ramaiah Institute of Technology [Link]

ACADEMIC AWARDS & FELLOWSHIPS

Graduate Fellowship, 2022	Awarded by the Dept. of CISE at University of Florida
Student Travel Grant, 2019	For attending CoDS-COMAD 2019

NATURAL LANGUAGES

English (native) · Kannada (native) · Hindi (working)